The Origin of Early Cosmic Structure and Ionizing Radiation: Quantifying the Role of Quasars via [O III] Emitters at $z \sim 7.5$

Abstract

Cosmic reionization transformed the intergalactic medium (IGM) from neutral to ionized at $z \sim 6$–20, coinciding with the emergence of the first supermassive black holes (SMBHs). A key unresolved question is how SMBHs were able to grow rapidly in this early epoch, and whether this growth is linked to overdense environments or rare accretion pathways. However, the relative roles of quasars and their surrounding environments in driving reionization and enabling early SMBH growth remain poorly constrained. Traditional tracers of ionizing sources, such as Lyα emission, are severely attenuated by the neutral IGM, limiting our ability to probe the surrounding galaxy population. We propose to conduct a systematic search for [O III] emitters around the $z = 7.5$ quasar J1007+2115 using JWST NIRCam imaging and grism spectroscopy. [O III] emission provides a robust and minimally attenuated tracer of ionized gas, enabling direct identification of ionizing sources in the reionization epoch. By identifying [O III] emitters through line excess in the F444W band, we will measure their spatial distribution and line fluxes. This characterization enables a quantitative mapping of the galaxy overdensity and the ionization structure of the quasar environment, directly testing whether early SMBH growth is associated with overdense regions or occurs in typical environments. This program establishes a direct observational link between SMBH growth and the reionization environment.

■ Scientific Justification

• Current observations cannot robustly determine whether z>7 quasars reside in overdense environments. The rapid growth of supermassive black holes (SMBHs) at z $\gg$ 7 remains one of the central challenges in early structure formation. Luminous quasars at these redshifts already host black holes with masses exceeding $10^9 M_\circ$ within the first < 1Gyr of cosmic history, implying extraordinarily efficient growth in the young Universe (Mortlock et al. 2011; Bañados et al. 2018; Wang et al. 2021). Meanwhile, standard Eddington-limited growth from stellar-mass seeds struggles to reproduce such extreme masses on the available timescale, motivating scenarios involving massive seeds, sustained super-Eddington accretion, or especially favorable environments (Volonteri 2012; Inayoshi et al. 2020). Existing observations further suggest that the environments of high-redshift quasars may not be uniform: some studies indicate overdense regions, while others are more consistent with typical field environments (Decarli et al. 2017; Overzier 2016). **Taken together, these results point to environment as a potentially crucial ingredient in early SMBH assembly, yet current constraints remain conflicting, making it essential to determine whether** z $\gg$ 7 **quasars preferentially inhabit overdense regions.**

However, a key limitation is **the lack of a robust census of galaxies in the vicinity environments of** z $\gg$ 7 **quasars**. Most previous searches have relied heavily on Ly α emission, but Ly α becomes increasingly unreliable in the reionization epoch because the neutral intergalactic medium strongly attenuates the line and suppresses its transmission (Stark et al. 2017; Matthee et al. 2018). In addition, interpreting Ly α-selected samples requires substantial modeling of radiative transfer and IGM transmission, making inferred galaxy densities highly model-dependent (Mason et al. 2018). As a result, present observations cannot yet distinguish whether rapid early SMBH growth is primarily enabled by overdense environments rich in nearby ionizing sources or by rare accretion pathways in otherwise typical regions. Without an unbiased tracer of galaxies in the reionization era, the connection between early quasar growth and local ionization structure remains unconstrained.

We propose to identify [O III] emitters around the z = 7.5 quasar J1007+2115 to construct the first spatially complete census of galaxies in its environment. JWST/NIRCam is uniquely suited to this measurement because it combines wide-field imaging, slitless grism spectroscopy, and access to rest-frame optical emission lines that are not strongly attenuated by the neutral IGM. Using NIRCam imaging and grism observations to identify [O III]-emitting galaxies, we can measure their spatial distribution and line fluxes and thereby map both the galaxy overdensity and the ionization structure in the quasar environment. These observations will provide the key empirical test of whether early SMBH growth is tied to dense large-scale environments or can proceed in more typical regions. **By delivering a benchmark dataset for quasar environments at $z > 7$, this program will directly advance models of SMBH formation, galaxy clustering, and reionization topology, while establishing a lasting legacy dataset for future JWST and Roman studies of the cosmic dawn.**

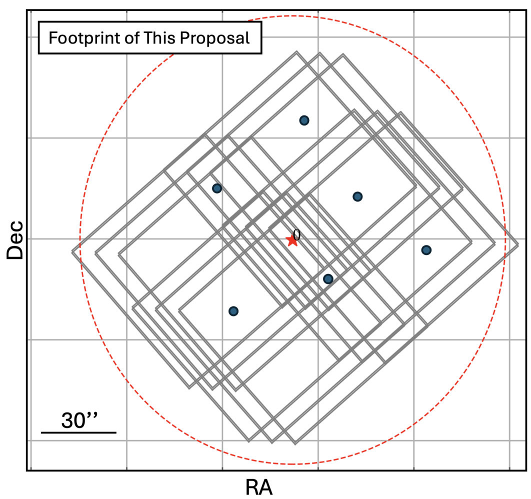
• **Current observations cannot robustly determine whether z > 7 quasars reside in overdense environments.** Observational studies of high-redshift quasar fields have yielded conflicting results, with some indicating significant galaxy overdensities while others are consistent with field-like environments. For example, searches for Ly α emitters around quasars have in some cases revealed enhanced galaxy populations (Decarli et al. 2017), while others find no statistically significant overdensity (Mazzucchelli et al. 2017). These discrepancies likely arise from differences in observational strategy, limited sample sizes, and the use of tracers strongly affected by the neutral intergalactic medium. **However, current observations cannot provide a robust and unbiased determination of the galaxy environments of z > 7 quasars.** The reliance on Ly α emission introduces severe, redshift-dependent selection biases, compounded by limited spatial coverage and incomplete spectroscopic confirmation, such that existing measurements cannot decisively distinguish between environment-driven and stochastic SMBH growth scenarios. **Resolving this discrepancy is essential for advancing the subfield**, as it directly constrains the physical conditions that enable rapid black hole growth during reionization.

The critical missing component is a **spatially complete and unbiased tracer of galaxies in quasar environments at z > 7**, minimally affected by IGM attenuation, with accurate redshift determination and wide-field applicability. Without this, a reliable census of galaxies and environmental overdensities remains impossible. This program will probe this missing component by identifying [O III] emitters around the **z = 7.5** quasar J1007+2115 using JWST/NIRCam imaging and grism spectroscopy. Because [O III] traces ionized gas and is largely unaffected by the neutral IGM, it provides a more robust probe of galaxy populations than Ly α. By delivering a spatially complete sample of [O III] emitters, this program will enable the first unbiased measurement of quasar environments at z > 7, providing the decisive test needed to resolve the origin of early SMBH growth.

• **This Proposal: Mapping [O III] Emitters Around a z = 7.5 Quasar with JWST/NIRCam Imaging and Slitless Spectroscopy.** We target the z = 7.5 quasar J1007+2115 and its surrounding galaxy population to probe the large-scale environment in which early supermassive black holes assemble. As one of the most distant known quasars, J1007+2115 provides a powerful laboratory for studying structure formation during the reionization epoch. The key requirement is to identify galaxies across a contiguous region with a tracer minimally affected by IGM attenuation. **JWST/NIRCam uniquely enables this program** by combining wide-field imaging and slitless grism spectroscopy with access to rest-frame optical emission lines such as [O III], which remain robust at z > 7. Unlike Ly α-based searches that suffer strong and spatially variable attenuation (Stark et al. 2017; Mason et al. 2018), NIRCam enables a uniform and physically meaningful census of the galaxy population.

We propose JWST/NIRCam imaging and grism spectroscopy of J1007+2115 to identify [O III] emitters and construct a spatially complete census of galaxies in its environment. Our strategy combines short-wavelength (SW) imaging and long-wavelength (LW) grism observations in a single, self-consistent setup. SW filters (F090W, F115W) will identify z $\gg$ 7 galaxies via Ly α dropout selection and provide photometric redshifts, while LW observations (F444W) detect [O III] emitters through line excess and confirm them via slitless spectroscopy. A dither pattern fills detector gaps to ensure continuous spatial coverage and enables high-quality imaging of the quasar host. As illustrated in Figure 1, this configuration provides uniform spatial sampling and efficient identification of emission-line galaxies, overcoming the limitations of Ly α-based approaches. Together, imaging and spectroscopy measure both the spatial distribution and line fluxes of [O III] emitters, enabling direct mapping of the ionized structure in the quasar environment.

The primary product of this program is a **spatially complete map of [O III] emitters around J1007+2115**. This dataset directly solves the central problem by providing the first unbiased measurement of galaxy overdensity and ionization structure in a $z > 7$ quasar environment, enabling a decisive test of whether early SMBH growth is linked to overdense regions.



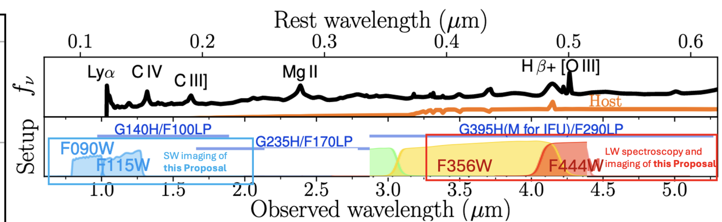


Figure 1 JWST/NIRCam will enable a contiguous census of [O III] emitters around a z=7.5 quasar, directly probing the role of environment in early SMBH growth. **Left**: proposed dither pattern, where the red star marks the quasar and blue points indicate expected [O III] emitters; the strategy fills detector gaps to achieve continuous spatial coverage and simultaneously probes the quasar host galaxy. **Right**: representative quasar and galaxy spectra with the adopted filter configuration, where SW filters identify the Lyα break via dropout selection and LW filters capture [O III] emission through line excess, while current Lyα-based approaches are limited by IGM attenuation and incomplete spatial sampling.
This program will map the spatial distribution of [O III] emitters around the quasar to test whether early SMBH growth is linked to overdense environments.

• **This program establishes a definitive observational link between early SMBH growth and the reionization environment.** The spatially complete census of [O III] emitters around a z¿7 quasar will provide the **first direct and unbiased measurement of galaxy overdensity and ionization structure in the immediate environments of early SMBHs**. These measurements will advance the subfield by decisively distinguishing between competing models of SMBH formation—environment-driven growth versus stochastic accretion—and by linking quasar activity to the topology of cosmic reionization.

This program is timely because JWST has only recently enabled access to rest-frame optical emission lines at z¿7, making it possible for the first time to probe galaxy environments with tracers that are not suppressed by the neutral IGM. This program directly addresses core JWST science goals of understanding the first galaxies, the sources of reionization, and the formation and growth of supermassive black holes in the early Universe. By delivering a benchmark dataset of [O III] emitters and their spatial distribution, this study will create a lasting legacy resource that enables cross-comparisons with simulations, informs future JWST and Roman surveys, and supports a broad range of investigations into galaxy clustering, ionization structure, and feedback during cosmic dawn.

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